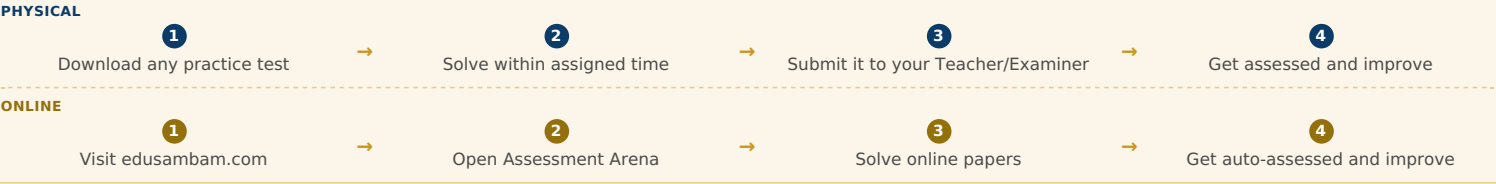


EDUSAMBAM ASSESSMENT SERIES

Algebra (O-Level Focus) & Graphs / Pie Charts
PRINTABLE PRACTICE WORKSHEET

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YOUR PHYSICAL & ONLINE SELF-ASSESSMENT ROADMAP



Candidate Name:		Roll No.:	
Class / Grade:		Date:	
School / Institution:		Test Centre:	

Subject	Mathematics (Maths)	Topic	Algebra (O-Level Focus) & Graphs / Pie Charts	Total Marks	100
Total Questions	45	Time Allowed	1 Hour (60 minutes)	Version	A

GENERAL INSTRUCTIONS

- Read each question carefully before answering.
- For Section A and Section B, mark one answer only.
- For Sections C and E, write your answer clearly in the space provided.
- For Section D, write the correct letter from Column B against each item in Column A.
- Do not use correction fluid. Keep all answers clear and legible.
- This worksheet is prepared from the EDUSAMBAM practice assessment supplied for this paper.

PAPER PATTERN

Section	Questions	Marks Each	Total
Section A — Multiple Choice	15	2	30
Section B — True / False	10	2	20
Section C — Fill in the Blank	10	2	20
Section D — Column Matching	5 pairs	2	10
Section E — Short Answer	5	4	20
Total	45	—	100

Section A — Multiple Choice

30 marks

1. Expand: $(x + 2)(x + 3)$
A. $x^2 + 5x + 6$
C. $x^2 + 5x + 5$
Answer: ☐A ☐B ☐C ☐D

B. $x^2 + 6x + 5$
D. $2x + 5$

Basic • 2 pts

2. Factorise: $x^2 + 7x + 12$
A. $(x+3)(x+4)$
C. $(x+1)(x+12)$
Answer: ☐A ☐B ☐C ☐D

B. $(x+2)(x+6)$
D. $(x+6)(x+6)$

Basic • 2 pts

3. Factorise: $x^2 - 9$
A. $(x-3)(x+3)$
C. $(x-3)^2$
Answer: ☐A ☐B ☐C ☐D

B. $(x-9)(x+1)$
D. $(x+9)(x-1)$

Basic • 2 pts

4. Solve the inequality: $x + 3 > 8$
A. $x > 5$
C. $x < 5$
Answer: ☐A ☐B ☐C ☐D

B. $x > 11$
D. $x > 3$

Basic • 2 pts

5. Simplify: $4x^2 \div 2x$
A. $2x$
C. $6x$
Answer: ☐A ☐B ☐C ☐D

B. $2x^2$
D. $8x$

Intermediate • 2 pts

6. If $y = 2x + 1$, what is y when $x = 3$?
A. 5
C. 7
Answer: ☐A ☐B ☐C ☐D

B. 6
D. 9

Intermediate • 2 pts

7. Which expression represents 'a number doubled, then increased by 5'?
A. $2x + 5$
C. $5x + 2$
Answer: ☐A ☐B ☐C ☐D

B. $x + 5 \times 2$
D. $2(x + 5)$

Intermediate • 2 pts

8. Solve: $2x - 3 = x + 5$
A. $x = 2$
C. $x = -8$
Answer: ☐A ☐B ☐C ☐D

B. $x = 8$
D. $x = 5$

Intermediate • 2 pts

9. A pie chart is best used to show:
A. Change over time
C. Exact individual values only
Answer: ☐A ☐B ☐C ☐D

B. Parts of a whole as percentages
D. Correlation between two variables

Intermediate • 2 pts

10. A full pie chart represents a total of:
A. 100 degrees
C. 180 degrees
Answer: ☐A ☐B ☐C ☐D

B. 360 degrees
D. 100 marks

Intermediate • 2 pts

11. If a pie chart slice takes up 25% of the circle, how many degrees is that?
A. 25°
C. 100°
Answer: ☐A ☐B ☐C ☐D

B. 90°
D. 360°

Intermediate • 2 pts

12. A line graph is most useful for showing:
A. Categories with no order
C. Parts of a whole
Answer: ☐A ☐B ☐C ☐D

B. Trends or change over time
D. A single value

Advanced • 2 pts

13. On a bar graph, the height (or length) of each bar represents:
A. The category name
C. The color
Answer: ☐A ☐B ☐C ☐D

B. The value or frequency
D. The x-axis label

Advanced • 2 pts

14. The horizontal axis of a graph is usually called the:
A. y-axis
C. z-axis
Answer: ☐A ☐B ☐C ☐D

B. x-axis
D. origin

Advanced • 2 pts

15. In a class of 40 students, 10 chose football as their favourite sport. What percentage is this?
A. 10%
C. 25%
Answer: ☐A ☐B ☐C ☐D

B. 20%
D. 40%

Advanced • 2 pts

Section B — True / False

20 marks

16. Factorising is the reverse process of expanding brackets.
A. True ☐ **B.** False ☐

Basic • 2 pts

17. $(x + 2)(x + 2)$ is the same as $x^2 + 4x + 4$.
A. True ☐ **B.** False ☐

Basic • 2 pts

18. The inequality $x > 5$ means x can equal exactly 5.
A. True ☐ **B.** False ☐

Basic • 2 pts

19. $x^2 - 4$ factorises to $(x-2)(x+2)$.
A. True ☐ **B.** False ☐

Basic • 2 pts

20. In algebra, 'simplify' and 'solve' mean exactly the same thing.
A. True ☐ **B.** False ☐

Intermediate • 2 pts

21. A pie chart always adds up to 360 degrees in total.
A. True ☐ **B.** False ☐

Intermediate • 2 pts

22. A larger slice of a pie chart always represents a larger percentage.
A. True ☐ **B.** False ☐

23. Bar graphs and line graphs use the same type of axes.
A. True ☐ **B.** False ☐

24. A line graph is the best choice for showing how temperature changes over a week.
A. True ☐ **B.** False ☐

25. Percentages in a pie chart must always add up to 100%.
A. True ☐ **B.** False ☐

Intermediate • 2 pts

Intermediate • 2 pts

Intermediate • 2 pts

Intermediate • 2 pts

Section C — Fill in the Blank

20 marks

26. Breaking an expression like $x^2 + 5x + 6$ into $(x+2)(x+3)$ is called _____.
Answer: _____

27. The opposite process of factorising is called _____.
Answer: _____

28. A mathematical sentence using $<$, $>$, $\leq$ or $\geq$ instead of an equals sign is called an _____.
Answer: _____

29. $(x + 5)(x + 5)$ can be written more shortly as $(x + 5)$ _____.
Answer: _____

30. $x^2 - 16$ factorises to $(x - 4)(x + \text{_____})$.
Answer: _____

31. A chart made of a full circle divided into slices is called a _____ chart.
Answer: _____

32. The vertical axis of a graph is usually called the _____-axis.
Answer: _____

33. A graph that uses connected points to show change over time is called a _____ graph.
Answer: _____

34. If a pie chart slice is 50% of the total, it takes up _____ degrees.
Answer: _____

35. A graph with separate rectangular columns is called a _____ graph or chart.
Answer: _____

Basic • 2 pts

Basic • 2 pts

Basic • 2 pts

Basic • 2 pts

Intermediate • 2 pts

Intermediate • 2 pts

Intermediate • 2 pts

Intermediate • 2 pts

Intermediate • 2 pts

Intermediate • 2 pts

Section D — Column Matching

10 marks

36-40. Match each term to its correct definition.

Column A	Answer	Column B
Factorise	_____	A. Multiply out brackets in an expression
Expand	_____	B. A statement comparing two expressions using $<$, $>$, $\leq$ or $\geq$
Inequality	_____	C. A graph showing change or trend over time using connected points
Pie chart	_____	D. Write an expression as a product of brackets
Line graph	_____	E. A circular chart divided into slices showing parts of a whole

Section E — Short Answer

20 marks

41. Explain what it means to factorise an algebraic expression.

42. Explain the difference between an equation and an inequality.

Advanced • 4 pts

Advanced • 4 pts

43. Describe how to expand $(x + 4)(x + 1)$.

Advanced · 4 pts

44. Explain when you would choose to use a pie chart instead of a bar graph.

Advanced · 4 pts

45. Describe how to work out how many degrees a slice of a pie chart should be if it represents 20% of the data.

Advanced · 4 pts

Candidate Signature: _____ **Invigilator Signature:** _____

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Marks Obtained: _____ / 100

Checked By: _____

Result / Grade: _____

Date Checked: _____

EDUSAMBAM ASSESSMENT SERIES

Algebra (O-Level Focus) & Graphs / Pie Charts — Answer Key

Teacher reference: equivalent grammatically correct wording is acceptable for open-ended items when the meaning is preserved.

Section A — Multiple Choice

- | | |
|------|-------|
| 1. A | 9. B |
| 2. A | 10. B |
| 3. A | 11. B |
| 4. A | 12. B |
| 5. A | 13. B |
| 6. C | 14. B |
| 7. A | 15. C |
| 8. B | |

Section B — True / False

- | | |
|-----------|----------|
| 16. True | 21. True |
| 17. True | 22. True |
| 18. False | 23. True |
| 19. True | 24. True |
| 20. False | 25. True |

Section C — Fill in the Blank

- | | |
|-----------------|----------|
| 26. factorising | 31. pie |
| 27. expanding | 32. y |
| 28. inequality | 33. line |
| 29. 2 | 34. 180 |
| 30. 4 | 35. bar |

Section D — Column Matching

- | | |
|--------------------|--------------------|
| 36. Factorise — D | 39. Pie chart — E |
| 37. Expand — A | 40. Line graph — C |
| 38. Inequality — B | |

Section E — Short Answer (Model Answers)

- 41.** brackets; product; reverse of expand
- 42.** equals; greater; less; range
- 43.** multiply; x squared; 5x; 4
- 44.** parts of a whole; percentage; proportion
- 45.** 360; 72; multiply; percentage